Question 3: Performance Comparison between Minimax and Alpha-Beta Pruning

In this specific game tree, both Minimax and Alpha-Beta pruning evaluate exactly 15 nodes. This includes 8 leaf nodes, 4 Level-2 MAX nodes (B, C, E, F), 2 Level-1 MIN nodes (A, D), and 1 ROOT node. No node savings occur because every internal node at Level 2 has exactly two leaf children. The pruning condition ($\alpha \geq \beta$) is only triggered after reading the second leaf of nodes C and F, meaning both leaves have already been visited and counted. For pruning to actually skip nodes, a node must have three or more children, allowing at least one entire subtree to be cut off without evaluation.